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Post-ERCP adverse events in children: a systematic review and meta-analysis of post-ERCP pancreatitis incidence with synthesis of risk factors and prevention strategies

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Abstract

Background Endoscopic retrograde cholangiopancreatography (ERCP) is increasingly used in children with pancreatobiliary disease, but pediatric adverse-event estimates remain variable. We updated the evidence on post-ERCP adverse events in children, pooled post-ERCP pancreatitis (PEP) incidence when study-level data allowed, and summarized reported risk factors and prevention strategies.

Methods We conducted a systematic review and meta-analysis of English-language literature published from January 2015 to June 2025 in PubMed, Embase, Web of Science, and the Cochrane Library. Eligible quantitative analyses included non-overlapping pediatric primary studies reporting extractable numerators and denominators for post-ERCP adverse events. Prior systematic reviews and meta-analyses were used for background comparison and reference checking only. Random-effects meta-analysis of proportions was planned for outcomes with at least three eligible studies. Risk-factor and prevention data were summarized narratively when definitions or adjustment strategies were too heterogeneous for pooling.

Results Sixty-five sources were included in the qualitative synthesis, and 15 non-overlapping primary studies contributed extractable PEP numerators and denominators for quantitative synthesis. PEP was the most frequent adverse event, with a pooled incidence of approximately 7.3% (95% CI, 5.1% to 10.4%; I-squared = 83.7%). Study-level PEP percentages, denominator units, and available adverse-event definition or reporting notes are now shown in Table 1. Bleeding was uncommon and perforation was rare but clinically important. Recurrent or chronic pancreatitis, pancreatic duct manipulation, pancreatic contrast injection, sphincterotomy, difficult cannulation, younger age, lower procedural volume, and trainee involvement or operator experience were recurring risk domains, although not all studies reported stratified event counts. Pediatric evidence for NSAIDs and prophylactic pancreatic stenting remains limited compared with adult data, and both strategies require child-specific dosing, equipment selection, and contraindication assessment.

Conclusions Pediatric ERCP carries measurable adverse-event risk, particularly PEP. The pooled estimate provides a current benchmark, but heterogeneity and inconsistent reporting limit certainty. Standardized pediatric definitions, non-overlapping reporting, and prospective multicenter datasets are needed.

Registration: PROSPERO CRD420251084264.

Keywords Children; ERCP; Pancreatobiliary disease; Post-procedural complications; Pancreatitis; Systematic review; Meta-analysis; Patient safety

Background

Endoscopic retrograde cholangiopancreatography (ERCP) has been used clinically since 1968 and is now a minimally invasive option for selected pancreatobiliary diseases[1]. In adults, it is widely used for common bile duct stones, biliary strictures, and pancreatitis. In children, ERCP is performed less often and requires attention to smaller anatomy, equipment constraints, and anesthesia risk[2-4]. Improvements in accessories and pediatric endoscopy expertise have expanded its use for choledocholithiasis, chronic pancreatitis, and post-liver transplant biliary complications[5-8].

Pediatric ERCP indications differ from adults and vary across age groups[4, 9]. For instance, choledocholithiasis is common in Western populations, whereas choledochal cysts are more prevalent in Asia[10]. ERCP may also be considered as an adjunct in selected diagnostic dilemmas such as neonatal cholestasis or suspected biliary atresia when non-invasive imaging is inconclusive, although its role must be balanced against procedural risk. Despite high technical success and clinical efficacy in children[11, 12], post-procedural complications remain an important clinical concern. The most common complications are post-ERCP

pancreatitis (PEP), bleeding, perforation, and infection[1]. These can prolong hospital stays, increase costs, and, in severe cases, threaten a child's life[13].

Previous research suggests that post-ERCP complication rates in children are broadly comparable to those in adults, but pediatric anatomy, indications, sedation requirements, and case volume create a distinct risk context. Compared with prior reviews and meta-analyses[4, 5, 13, 14], the present review updates the evidence through June 2025, separates primary-study data from prior pooled estimates, avoids double counting overlapping cohorts, and restricts quantitative synthesis to outcomes with extractable non-overlapping data.

Methods

This systematic review and meta-analysis was reported in accordance with PRISMA 2020[15]. The protocol was prospectively registered in PROSPERO (CRD420251084264), in accordance with best practice guidelines to ensure transparent reporting and reduce reporting bias.

Search Strategy

We performed a systematic review and meta-analysis to identify, pool, and stratify reported post-ERCP adverse events in pediatric patients. Two reviewers independently searched PubMed, Embase, Web of Science, and the Cochrane Library for literature on pediatric ERCP complications. Any discrepancies in study selection were resolved through discussion or consultation with a third reviewer. The search strategy combined Medical Subject Headings (MeSH) terms and free-text keywords using Boolean operators (AND, OR). Keywords included: “children”, “pediatric”, “ERCP”, “endoscopic retrograde cholangiopancreatography”, “complications”, “adverse event”, “pancreatitis”, “bleeding”, “perforation”, and “infection”.

The search was limited to studies published between January 1, 2015, and June 26, 2025, to capture contemporary pediatric ERCP practice, including changes in therapeutic indications, endoscopic accessories, anesthesia support, and prophylaxis strategies. Earlier evidence was considered through prior reviews only for contextual comparison. Reference lists of eligible studies and relevant reviews were manually checked to identify additional contemporary primary studies.

Inclusion and Exclusion Criteria

Studies were selected using predefined criteria:

Inclusion Criteria:

Study Type: Original pediatric studies, including cohort studies, case-control studies, and randomized or non-randomized interventional studies. Prior systematic reviews and meta-analyses were screened for contextual comparison and citation tracking only. Their pooled results were not entered into the present quantitative synthesis.

Population: Pediatric patients (<18 years) undergoing ERCP.

Outcomes: Reported incidence, types, risk factors, or management strategies of post-ERCP complications.

Language: English.

Publication Date: January 1, 2015 to June 26, 2025.

Exclusion Criteria:

Population: Adult patients (≥ 18 years).

Intervention: Research unrelated to ERCP complications.

Study Design: Case reports, narrative reviews, editorials, conference abstracts, animal studies, in vitro studies.

Data Availability: Incomplete or unextractable data.

Duplication: Repeated publications.

Data Extraction

Data extraction was independently conducted by two reviewers using a standardized form. Any discrepancies were resolved through discussion or consultation with a third reviewer. For each primary study, we collected first author, year, country, study design, sample size, number of ERCP procedures, mean or median age, ERCP indications, procedure type, follow-up duration, adverse-event definitions or reporting windows, event numerators and denominators for PEP, bleeding, perforation, cholangitis or infection, severity classification, risk factors, and management strategies. For PEP, we recorded whether the denominator represented ERCP procedures, patients, first ERCP per child, or a mixed/unclear reporting unit. We retained the denominator unit used by the source rather than converting procedure-level data to patient-level data when repeat procedures per child were not consistently reported. For prior systematic reviews and meta-analyses, we extracted scope, search period, included study types, pooled estimates, and overlap with primary studies to avoid

double counting.

Quality Assessment

The methodological quality and risk of bias of each eligible source were independently assessed by two reviewers using tools appropriate to study design. Disagreements were resolved by consensus, with a third reviewer consulted when needed.

Non-randomized primary studies were assessed using the Newcastle-Ottawa Scale (NOS), randomized trials using RoB 2, and prior systematic reviews or meta-analyses using AMSTAR 2[16-18]. Rather than reproducing checklist content, we summarized each study's overall judgment and used these judgments to grade confidence across evidence domains.

Risk-of-bias judgments informed the narrative synthesis. Studies with incomplete adverse-event definitions, unclear denominators, overlapping cohorts, or high risk of selection or reporting bias were interpreted cautiously and were not used to support quantitative claims beyond their reported data.

Statistical Analysis

Meta-analysis was conducted only for outcomes reported by at least three non-overlapping primary studies with extractable numerators and denominators. Pooled incidences were estimated using random-effects models for proportions, with variance stabilization for rare events where appropriate. Results were expressed as pooled proportions with 95% confidence intervals (CIs).

Between-study heterogeneity was assessed using Cochran Q and the I-squared statistic. Heterogeneity was interpreted as low, moderate, substantial, or considerable using conventional thresholds, while acknowledging that I-squared may be unstable in rare-event analyses. We prespecified subgroup exploration by age group, ERCP indication, procedure type, pancreatic duct manipulation or contrast injection, sphincterotomy, and operator or center volume. Formal subgroup meta-analysis was not forced when studies did not provide stratified PEP event counts, because doing so would have required ecological assumptions. These variables were therefore summarized as explanatory domains for heterogeneity and risk.

Sensitivity analyses were planned by excluding studies at high risk of bias, studies with unclear adverse-event definitions, and studies with potential cohort overlap. In response to peer review, we also added a post hoc denominator-unit sensitivity summary, separating procedure-level studies from patient-level or first-ERCP-per-child studies and retaining a mixed/unclear category when the source did not permit classification. Risk-factor effect estimates were not pooled when exposure definitions, covariate adjustment, or reporting units differed substantially. These data were summarized narratively and, where appropriate, displayed as reported estimates.

Publication bias was assessed visually using funnel plots and statistically using Egger test only for outcomes with at least ten eligible studies. Funnel plots were not generated for rare outcomes with fewer than ten studies because asymmetry tests would be unreliable.

All tests were two-sided. For analyses other than Cochran's Q test for heterogeneity, $P < 0.05$ was considered statistically significant. For Cochran's Q test, $P < 0.10$ was considered indicative of potential between-study heterogeneity because this test has limited power when relatively few studies are included. Heterogeneity was interpreted together with I^2 and τ^2 .

AI-assisted tools were used during revision for language editing and formatting checks. The authors verified the data extraction, statistical analyses, interpretation, and final text and take full responsibility for the manuscript.

Results

This systematic review included 65 sources in the qualitative synthesis, consisting primarily of contemporary pediatric ERCP primary studies and selected prior systematic reviews or meta-analyses used for contextual benchmarking[2-14, 19-71]. Prior reviews were not treated as independent primary data and their pooled estimates were not combined with overlapping primary studies. A detailed account of the literature screening and selection process, following PRISMA 2020 principles, is illustrated in Figure 1.

Characteristics of Included Studies

This review included contemporary pediatric ERCP primary studies and selected prior reviews used for contextual benchmarking. Table 1 summarizes the primary studies that contributed PEP event data to the quantitative synthesis and identifies prior systematic reviews used only for comparison or citation tracking. This

separation reduces double counting of overlapping evidence.

Risk-of-Bias Summary

Most included primary studies were retrospective or observational, and the overall certainty of evidence was limited by heterogeneous adverse-event definitions, variable follow-up windows, incomplete reporting of denominators for specific complications, and potential overlap among multicenter pediatric ERCP cohorts. Contemporary multicenter studies generally provided stronger evidence for technical outcomes and common adverse events, whereas estimates for rare events such as perforation and cholangitis were less precise. Prior systematic reviews and meta-analyses were useful for benchmarking but were not used as data sources in the present pooled analyses.

Incidence of complications

The available evidence suggested that ERCP indication influenced complication patterns. Children undergoing ERCP for acute recurrent or chronic pancreatitis were repeatedly reported to have higher PEP risk than those undergoing ERCP for choledocholithiasis, although the magnitude of association varied by study design, adjustment strategy, and procedural case mix.

The overall incidence of post-ERCP complications in children varied across studies because of differences in patient populations, procedural indications, adverse-event definitions, and follow-up. Random-effects pooling was performed for PEP because this outcome had the most consistently extractable numerators and denominators. Figure 2 presents the study-level forest plot for PEP incidence after pediatric ERCP. Bleeding, perforation, and cholangitis were less frequent and were summarized descriptively when study-level event data were sparse or inconsistently reported.

PEP: PEP was the most frequent and clinically significant complication associated with pediatric ERCP. In the present synthesis, 15 non-overlapping primary studies contributed extractable PEP event counts, yielding a pooled incidence of 7.3% (95% CI, 5.1% to 10.4%; I-squared = 83.7%) under a random-effects model. Table 1 now reports the study-level numerator, denominator, percentage, denominator unit, and available definition or reporting note for every study contributing to the PEP synthesis. The rows most dependent on percentage back-calculation or source-unit interpretation were rechecked against the original source records for Goetz, Mercier, Perera, and Batibay[24, 60, 63, 72].

The high I-squared value indicates real between-study inconsistency rather than a single uniform pediatric PEP risk. Likely contributors include age distribution, acute recurrent or chronic pancreatitis as the indication, pancreatic duct cannulation or contrast injection, pancreatic sphincterotomy, difficult cannulation, operator and center volume, trainee participation, follow-up windows, and differences in PEP definitions. Denominator unit was another source of variation: some studies reported ERCP procedures, whereas others reported patients or first ERCP per child. Because repeat ERCPs per child were not consistently available, procedure-level data were not converted to patient-level data. In the post hoc denominator-unit summary, procedure-level studies contributed 181 PEP events among 2669 ERCPs (crude rate, 6.8%), patient-level or first-ERCP-per-child studies contributed 20 events among 421 children (4.8%), and the mixed or unclear-denominator study contributed 12 events among 232 observations (5.2%). Heterogeneity persisted after this separation, so the pooled estimate should be interpreted as a contemporary benchmark rather than a single individualized risk estimate.

Most pediatric PEP cases are mild and resolve with conservative management, whereas a minority progress to moderate or severe PEP with prolonged hospitalization, systemic inflammatory response, pancreatic necrosis, pseudocyst formation, or rarely mortality[4, 59, 73]. The Atlanta classification, originally developed for adults, is often applied in pediatric settings but has not been fully validated for pediatric post-ERCP pathophysiology [62].

Visual inspection of the PEP funnel plot suggested possible small-study effects, although Egger regression did not show clear statistical evidence of asymmetry (intercept, -1.98; approximate two-sided P = 0.16). This result should be interpreted cautiously because heterogeneity was substantial and denominators were reported as either patients or procedures across studies (Figure 8).

Bleeding: Post-ERCP bleeding in children most commonly occurs after endoscopic sphincterotomy (EST).

Clinically significant bleeding is uncommon, with an estimated incidence of approximately 0.5%[4]. Bleeding may appear during the procedure, shortly afterward, or several days later. Reported risk factors include coagulopathy, larger sphincterotomy incisions, and vascular anomalies. Most cases respond to endoscopic hemostasis with epinephrine injection or clip placement. Severe or endoscopically refractory bleeding may require angiographic embolization; surgery should be reserved for cases in which less invasive measures fail.

Perforation: ERCP-related perforation is rare but potentially life-threatening, with an estimated incidence below 1% in pediatric patients[4]. Risk factors include difficult cannulation, excessive force during guidewire or endoscope advancement, anatomical anomalies, and strictures. A practical concern is device size. Pediatric duodenoscopes have become less available in recent years, so clinicians may need to use adult-sized equipment in small children, which can increase mechanical risk. Early recognition matters: minor perforations may be treated conservatively, whereas larger defects often require surgery.

Cholangitis: Post-ERCP cholangitis is a recognized infectious complication, typically resulting from incomplete biliary drainage or procedural contamination. Although its incidence in children is less well-defined than that of PEP, it remains clinically important[63]. Risk factors include persistent biliary obstruction, incomplete stone clearance, and prior biliary surgery. Prophylactic antibiotics are commonly considered, particularly in patients with pre-existing cholangitis or immunocompromised status. Management requires prompt antibiotic therapy and restoration of biliary drainage, which may require repeat ERCP or percutaneous intervention.

Other complications: Less common complications of ERCP in children include cholecystitis, bile duct injury or stricture, and events related to sedation or anesthesia[13]. Cholecystitis may arise if the cystic duct is obstructed or manipulated during the procedure. Ductal injury is rare but can lead to later stricture formation. Sedation- and anesthesia-related complications are especially relevant in children because of smaller airway size, higher metabolic demand, and age-specific responses to anesthetic agents. Close monitoring by pediatric anesthesia teams is therefore important for procedural safety[64].

The evidence points to a broad but unevenly reported risk profile for pediatric ERCP. Overall complication rates are usually low, but serious events such as PEP, bleeding, perforation, and cholangitis require close attention. Differences in definitions, follow-up, and reporting explain much of the variability across studies. Better pediatric definitions and prospective datasets are needed before risk estimates for rare outcomes can be pooled with confidence.

Risk Factors for Complications

Operator background also influenced complication rates. Pediatric series and adult operator-volume data[66, 68, 69] suggest that procedures performed by less experienced operators or outside specialized pediatric endoscopy settings may carry higher procedural risk than procedures performed by experienced pediatric gastroenterologists. These findings support specialized training and referral pathways for complex pediatric ERCP.

Identifying risk factors for post-ERCP complications in children helps guide patient selection, pre-procedural optimization, and intra-procedural risk mitigation. Across the literature, both patient characteristics and procedural variables were associated with adverse events.

Table 2 summarizes major patient- and procedure-related factors associated with higher complication rates, particularly PEP, bleeding, and perforation.

To complement this summary, Figure 3 summarizes patient-related and procedure-related risk domains reported across the included literature.

Patient-Related Risk Factors

Age: Younger age, particularly infancy and early childhood, has been reported as a possible risk factor for PEP in several pediatric cohorts[65]. Smaller ductal anatomy, pancreatic reactivity, and age-specific inflammatory responses may contribute. However, some multivariable analyses suggest that age may not remain an independent predictor after adjustment for indication and procedural factors. Figure 4 illustrates the reported age pattern, but the finding should be interpreted as hypothesis-generating rather than definitive.

History of Pancreatitis: A prior diagnosis of acute recurrent pancreatitis or chronic pancreatitis is strongly associated with an elevated risk of PEP in children undergoing ERCP[65]. This increased susceptibility likely reflects underlying pancreatic ductal distortion, ongoing low-grade inflammation, and fibrosis, all of which may amplify the pancreas's vulnerability to procedural trauma.

Sphincter of Oddi Dysfunction (SOD): Although SOD is more commonly recognized in adults, pediatric

patients with suspected or confirmed SOD appear to face a greater risk of PEP. This may stem from impaired pancreatic outflow dynamics and elevated ductal pressures, which exacerbate the likelihood of post-procedural inflammation[65, 66].

Genetic Predisposition: Variants linked to hereditary pancreatitis, including PRSS1 and SPINK1, may increase baseline susceptibility to pancreatic injury[67]. Their relevance to post-ERCP risk remains uncertain, but their presence may help identify children who require closer procedural planning and follow-up.

Procedure-Related Risk Factors

Pancreatic Duct Cannulation and Sphincterotomy: Pancreatic duct instrumentation, including deep cannulation, repeated cannulation attempts, and pancreatic sphincterotomy, has been associated with higher PEP risk in pediatric cohorts[65]. These findings parallel adult multicenter risk-factor data and support minimizing pancreatic manipulation unless it is clinically necessary[74, 75]. Selective biliary cannulation remains an important risk-reduction strategy.

Precut Papillotomy: Precut papillotomy, often reserved for cases of difficult biliary access, carries an increased risk of both PEP and duodenal perforation due to greater mechanical trauma to the ampullary complex[68]. Although it can facilitate access in anatomically challenging cases, its use in children should be judicious and ideally limited to experienced operators.

Pancreatic Stent Placement: While prophylactic pancreatic stent placement has demonstrated efficacy in reducing PEP in high-risk adult populations, its role in pediatric ERCP remains less well-defined. Improperly selected or malpositioned stents may paradoxically increase the risk of ductal injury, stent-induced pancreatitis, or migration-related complications[65]. Pediatric-specific evidence is limited, and further work is needed to define appropriate stent design and placement protocols in children.

Operator Experience: Lower operator or center volume and trainee involvement may contribute to adverse events, particularly during difficult cannulation or pancreatic duct manipulation. The available pediatric studies do not provide enough stratified event counts for formal pooling by operator experience, but the recurring association supports referral of complex pediatric ERCP to experienced teams and high-volume centers[76].

Procedure Complexity and Duration: Complex procedures, including those involving multiple therapeutic steps, altered anatomy, or prolonged duration, are associated with higher adverse-event risk[68]. These factors probably reflect both the difficulty of the underlying case and cumulative tissue trauma.

Where sufficiently comparable adjusted estimates were available, risk-factor associations were extracted. However, heterogeneity in exposure definitions and covariate adjustment generally precluded formal pooling. Figure 5 therefore presents selected reported effect estimates for key PEP predictors and should be interpreted as a structured risk-factor display rather than a pooled predictor meta-analysis.

Management and Prevention Strategies

Improving pediatric ERCP safety requires attention before, during, and after the procedure. Practical countermeasures include risk stratification, avoidance of unnecessary pancreatic duct trauma, selective rather than routine prophylaxis, child-sized stent selection when stenting is used, careful NSAID eligibility assessment, and structured post-procedural monitoring. Table 3 summarizes prevention and management strategies for the main ERCP-related complications in children.

Prevention of PEP

Prevention of PEP remains a central objective in pediatric ERCP practice. Strategies adapted from adult guideline recommendations and limited pediatric data have potential value, but pediatric use requires attention to anatomy, body size, renal function, bleeding risk, and local expertise[71, 77].

Patient Selection: Careful pre-procedural evaluation is essential. The decision to proceed with ERCP should be based on risk-benefit assessment. Children with recurrent pancreatitis or anticipated complex pancreatic interventions should be categorized as high risk and managed accordingly.

Prophylactic Pancreatic Stent Placement: Prophylactic pancreatic stenting is widely used in high-risk adult ERCP and may reduce PEP risk in selected pediatric cases, particularly after pancreatic sphincterotomy or difficult pancreatic duct cannulation[65, 70, 77]. In children, the decision is not simply whether to place a stent, but whether the duct size, papillary anatomy, and operator experience allow safe placement. Stent caliber, length, flange design, and planned dwell time should be matched to the child and the pancreatic duct; adult-sized

assumptions may increase ductal trauma, migration, or stent-induced pancreatitis. Current pediatric data are insufficient to recommend one universal stent protocol.

Nonsteroidal Anti-inflammatory Drugs (NSAIDs): Rectal indomethacin is established for PEP prophylaxis in adults, but pediatric evidence remains limited and should not be treated as a direct adult-to-child transfer[78, 79]. Pediatric use should account for age, body weight, renal function, hydration status, bleeding risk, allergy history, and institutional protocols. The available pediatric feasibility work, including intravenous ibuprofen, supports further study but does not define a universal dose or timing strategy for all children[71].

Aggressive Intravenous Hydration: Periprocedural hydration with lactated Ringer solution reduces PEP risk in adults[80]. In children, this approach requires careful fluid monitoring to avoid overload, especially in younger patients or those with comorbid disease.

Minimizing Pancreatic Duct Manipulation: Limiting unnecessary pancreatic duct instrumentation is critical. Strategies such as reducing the number of cannulation attempts, avoiding deep cannulation when not clinically indicated, and using wire-guided cannulation techniques have been associated with lower PEP rates by minimizing ductal trauma[68].

Operator Experience: Endoscopist experience is a consistent determinant of procedural safety. ERCPs performed in high-volume centers by experienced operators are associated with lower PEP and overall complication rates[76]. When trainees participate, supervision should include predefined limits for cannulation attempts, early senior-operator takeover during difficult access, and avoidance of unnecessary pancreatic duct injection. Figure 6 summarizes preventive strategies according to clinical utility and maturity of supporting evidence.

Management of Post-ERCP Complications

The management of ERCP-related complications in children is guided by the nature and severity of each event:

PEP: Mild to moderate cases are typically managed conservatively with bowel rest when clinically indicated, intravenous fluids, analgesia, and antiemetics. Early enteral nutrition should be considered as recovery permits. In severe PEP, additional interventions may include antibiotics for infected necrosis, image-guided percutaneous or endoscopic drainage of fluid collections, and, rarely, surgical debridement[81]. Figure 7 provides a simplified severity and management framework, because it is adapted largely from adult-based Atlanta concepts, it should be used pragmatically and not as a fully validated pediatric-specific severity instrument.

Bleeding: Most episodes of post-ERCP bleeding are self-limiting. Clinically significant hemorrhage requires prompt endoscopic treatment, including epinephrine injection, hemostatic clip placement, or thermal coagulation[82]. Blood transfusion may be necessary in patients with hemodynamic instability. Refractory bleeding may require angiographic embolization or surgical control[83].

Perforation: Management depends on perforation type, location, and clinical impact. Minor retroperitoneal perforations, often detected by retroperitoneal air on imaging, may be managed with bowel rest, intravenous fluids, antibiotics, and close observation[84]. Selected defects may be closed endoscopically with clips or stents. Large perforations, especially those with free intraperitoneal air or peritonitis, usually require urgent surgical repair[85].

Cholangitis: Post-ERCP cholangitis requires immediate broad-spectrum empirical antibiotics targeting common biliary pathogens. Definitive management necessitates biliary drainage restoration, preferably via endoscopic stenting or stone extraction; surgical drainage should be considered when endoscopic intervention fails or is contraindicated. Timely management is essential to prevent sepsis and other severe complications[86, 87].

Discussion

Our analysis also identified features associated with severe complications requiring surgical intervention, including prior pancreatitis, prolonged procedure duration, and repeated pancreatic duct cannulation attempts. Children differ from adults not only in anatomy and duct size but also in how complications may present. Children with low body weight may deteriorate faster from fluid shifts or infection, whereas adult risk profiles are more often shaped by cardiovascular comorbidity. Pediatric ERCP therefore requires age-specific risk assessment rather than direct extrapolation from adult practice.

This systematic review and meta-analysis synthesizes contemporary evidence on post-ERCP complications in children. Prior reviews were used for context, while original pediatric studies served as the primary evidence base. The findings support ERCP as an important diagnostic and therapeutic tool in pediatric pancreatobiliary disease, but they also show that PEP remains the main adverse event requiring prevention, early recognition, and standardized reporting.

Consistent with prior reports, PEP was the most commonly reported adverse event. The present pooled estimate was 7.3% (95% CI, 5.1% to 10.4%), with substantial between-study heterogeneity ($I^2 = 83.7\%$). Table 4 compares this estimate with prior pediatric ERCP meta-analyses. Usatin 2016 and Hosseini 2023 reported lower PEP estimates of approximately 3% to 4%, whereas Sun 2022 provided an overall therapeutic ERCP adverse-event benchmark rather than a uniformly extractable PEP-only estimate[4, 5, 13]. The higher estimate in the present review is likely related to the contemporary search window, explicit restriction to non-overlapping primary studies with extractable PEP numerators and denominators, high-risk therapeutic indications, pancreatic duct manipulation, and retention of the source-specific denominator unit. Standardized adverse-event reporting using established classifications (e.g., Cotton criteria[88]; revised Atlanta criteria[89]) and pediatric-specific definitions is needed for valid cross-study comparisons.

Although PEP was the most commonly reported adverse event, less frequent complications such as bleeding and perforation also require attention. Bleeding was generally uncommon (approximately 0.5%) and was most often associated with endoscopic sphincterotomy. Perforation was rare, generally reported below 1%, but its severity justifies careful technique, especially in children with smaller and more delicate duodenal and biliary anatomy.

Risk reduction depends on both patient selection and procedural technique. Younger age, prior pancreatitis, and selected genetic predispositions were recurring patient-level risk factors. Procedural factors such as pancreatic duct manipulation (including deep cannulation and pancreatic sphincterotomy), precut papillotomy, and lower operator experience were also associated with higher risk. Individual risk stratification and referral to high-volume pediatric endoscopy centers may reduce avoidable harm[66].

Preventive strategies remain central to reducing post-ERCP morbidity, but pediatric practice requires more detail than simply importing adult ERCP protocols. For PEP, the most actionable measures are careful indication selection, minimizing pancreatic duct manipulation and contrast injection, limiting cannulation attempts, early senior-operator involvement, and selective prophylactic pancreatic stenting when anatomy and expertise make it safe. Stent size and length should be adapted to the child and duct, and NSAID prophylaxis should be individualized by age, weight, renal function, bleeding risk, and local pediatric protocols. For bleeding, perforation, and cholangitis, countermeasures include correction of coagulation abnormalities, careful sphincterotomy, early recognition of perforation, complete biliary drainage, and selective antibiotics when drainage is incomplete or infection risk is high.

Selective prophylactic pancreatic stenting may benefit high-risk pediatric cases, but the evidence remains limited. Intravenous hydration may also help when fluid status can be monitored closely[65, 80].

This study has several strengths: a contemporary search strategy, explicit separation of primary studies from prior reviews, rechecking of percentage-derived PEP rows, and a meta-analytic approach restricted to outcomes with extractable non-overlapping data. Limitations remain. Heterogeneity in indications, adverse-event definitions, follow-up windows, and reporting units limited comparison across studies and prevented pooling of several risk-factor and prevention outcomes. Some studies used procedure-level denominators whereas others used patient-level or first-ERCP-per-child denominators; because repeat procedures per child were not consistently reported, these could not be fully harmonized. Selective reporting, incomplete stratified event counts, and overlap among multicenter cohorts may also affect published estimates.

Future research should prioritize prospective multicenter pediatric ERCP datasets with standardized adverse-event definitions and consistent reporting of numerators, denominators, follow-up windows, and severity. Pediatric trials or well-designed registries are also needed for prophylactic agents, pancreatic stenting strategies, and child-adapted ERCP equipment.

Conclusions

Pediatric ERCP is effective, but adverse events, especially PEP, remain clinically important. A focused meta-analysis of extractable PEP data provides a transparent incidence estimate, while the broader evidence synthesis shows where pediatric data are still thin. Risk reduction in pediatric ERCP will depend on careful patient selection, experienced operators, standardized reporting, and prospective pediatric evidence for prophylaxis.

Abbreviations

AMSTAR 2: A Measurement Tool to Assess Systematic Reviews 2; CI: confidence interval; ERCP: endoscopic retrograde cholangiopancreatography; EST: endoscopic sphincterotomy; NSAIDs: nonsteroidal anti-inflammatory drugs; PEP: post-ERCP pancreatitis; PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PROSPERO: International Prospective Register of Systematic Reviews.

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Authors' contributions

Xin Wang and Hua Liu made the largest contributions to this work. Xin Wang conceived the review question, designed the search and extraction framework, reconstructed the study-level dataset, performed the meta-analysis, generated the figures, and drafted the manuscript. Ruiying Li contributed to literature screening, data extraction, and table preparation. Wenhai Li contributed to source checking, data verification, and synthesis of risk-factor evidence. Peng Huang contributed to statistical checking, figure review, and interpretation of pooled estimates. Hu Yang contributed to clinical interpretation of pediatric surgical and endoscopic findings and revised the manuscript for important intellectual content. Hongxi Guo contributed to quality assessment, supplementary file preparation, and final source verification. Hua Liu supervised the study, refined the study design, resolved disagreements, interpreted the clinical implications, and critically revised the manuscript. All authors read and approved the final manuscript.

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Availability of data and materials

The reconstructed extraction sheet, verified PEP dataset, analysis script, and figure source data are provided as supplementary files. Additional details are available from the corresponding author on reasonable request.

Ethics declarations

Ethics approval and consent to participate

Not applicable. This study is a systematic review and meta-analysis of published data and did not involve new recruitment of human participants or access to identifiable individual patient data.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

References

1. Kozarek RA. The Past, Present, and Future of Endoscopic Retrograde Cholangiopancreatography. *Gastroenterol Hepatol (N Y)*. 2017; 13(10): 620-622.
2. Lorio E, Moreau C, Hernandez B, et al. Pediatric ERCP: Factors for Success and Complication-A 17-Year, Multisite Experience. *J Pediatr Gastroenterol Nutr*. 2023;77(3):413-21.
3. Çirkin G, Akarsu M, Öztürk Y, İyilikçi L, Güler Y, Gülpınar AÖ. Evaluation of endoscopic retrograde cholangiopancreatography in Turkish children. *Medicine (Baltimore)*. 2024;103(52):e41045.
4. Hosseini A, Sohoulı MH, Sharifi E, Sayyari A, Sridharan K, Tajalli S, Imanzadeh N, Fatahi S. Indications, success, and adverse event rates of pediatric endoscopic retrograde cholangiopancreatography (ERCP): a systematic review and meta-analysis. *BMC Pediatr*. 2023;23(1):596.
5. Sun R, Xu X, Zheng Q, Zhan J. Therapeutic Endoscopic Retrograde Cholangiopancreatography for Pediatric Hepato-Pancreato-Biliary Diseases: A Systematic Review and Meta-Analysis. *Front Pediatr*. 2022;10:915085.
6. Liu QY, Gugig R, Troendle DM, et al. The Roles of Endoscopic Ultrasound and Endoscopic Retrograde Cholangiopancreatography in the Evaluation and Treatment of Chronic Pancreatitis in Children: A Position Paper From the North American Society for Pediatric Gastroenterology, Hepatology, and Nutrition Pancreas Committee. *J Pediatr Gastroenterol Nutr*. 2020;70(5):681-93.
7. Dişçi E, Peksöz R, Yıldız M, Yıldırğan Mİ, Albayrak Y, Fakirullahoğlu M, Fırıncı B, Atamanalp SS. Endoscopic Retrograde

- Cholangiopancreatography in Pediatric Patients. *J Laparoendosc Adv Surg Tech A*. 2022;32(3):320-4.
8. Dechène A, Kodde C, Kathemann S, et al. Endoscopic treatment of pediatric post-transplant biliary complications is safe and effective. *Dig Endosc*. 2015;27(4):505-11.
 9. Keil R, Drábek J, Lochmannová J, et al. ERCP in infants, children, and adolescents-Different roles of the methods in different age groups. *PLoS One*. 2019;14(1):e0210805.
 10. Baison GN, Bonds MM, Helton WS, Kozarek RA. Choledochal cysts: Similarities and differences between Asian and Western countries. *World J Gastroenterol*. 2019;25(26):3334-43.
 11. Giefer MJ, Kozarek RA. Technical outcomes and complications of pediatric ERCP. *Surg Endosc*. 2015;29(12):3543-50.
 12. Poddar U, Samanta A, Mohindra S, Upadhyaya VD, Kumar B, Srivastava A, Sen Sarma M, Yachha SK. Endoscopic retrograde cholangiopancreatography and endoscopic cystogastrostomy in very young children (aged <5 years): Feasibility, success, and safety. *DEN Open*. 2025;5(1):e70085.
 13. Usatin D, Fernandes M, Allen IE, Perito ER, Ostroff J, Heyman MB. Complications of Endoscopic Retrograde Cholangiopancreatography in Pediatric Patients; A Systematic Literature Review and Meta-Analysis. *J Pediatr*. 2016;179:160-5.e3.
 14. Barakat M, Saumoy M, Forbes N, Elmunzer BJ. Complications of Endoscopic Retrograde Cholangiopancreatography. *Gastroenterology*. 2025;169(2):230-43.e8.
 15. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.
 16. Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. *Eur J Epidemiol*. 2010;25(9):603-5.
 17. Sterne J, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898.
 18. Shea BJ, Reeves BC, Wells G, et al. AMSTAR 2: a critical appraisal tool for systematic reviews that include randomised or non-randomised studies of healthcare interventions, or both. *BMJ*. 2017;358:j4008.
 19. Deng Z, Zeng J, Lv C, Jiang L, Ji J, Li X, Hao L, Gong B. Prevalence and Factors Associated with Post-Endoscopic Retrograde Cholangiopancreatography Pancreatitis in Children. *Dig Dis Sci*. 2021;66(1):224-30.
 20. Zhu R, Wu S, Xu J, Wang R, Cheng Y, Xu Y, Chen C, Wu C. Temporal trend and factors associated with post-endoscopic retrograde cholangiopancreatography pancreatitis in children in the USA: a nationwide, retrospective cohort study. *Eur J Pediatr*. 2023;182(5):2273-82.
 21. ASGE Standards of Practice Committee, Chandrasekhara V, Khashab MA, et al. Adverse events associated with ERCP. *Gastrointest Endosc*. 2017;85(1):32-47.
 22. Ribeiro IB, do Monte Junior ES, Miranda Neto AA, et al. Pancreatitis after endoscopic retrograde cholangiopancreatography: A narrative review. *World J Gastroenterol*. 2021;27(20):2495-506.
 23. Troendle DM, Ruan W, Fishman DS, et al. Technical Outcomes in Pediatric Endoscopic Retrograde Cholangiopancreatography: Data from an International Collaborative. *J Pediatr Gastroenterol Nutr*. 2022;75(6):755-60.
 24. Mercier C, Pioche M, Albuisson E, et al. Safety of endoscopic retrograde cholangiopancreatography in the pediatric population: a multicenter study. *Endoscopy*. 2021;53(6):586-94.
 25. Trocchia C, Khalaf R, Amankwah E, et al. Pediatric ERCP in the Setting of Acute Pancreatitis: A Secondary Analysis of an International Multicenter Cohort Study. *J Pediatr Gastroenterol Nutr*. 2023;76(6):817-21.
 26. Shah R, Cohen RZ, Mekaroonkamol P, et al. Retrospective Multicenter Matched Controlled Comparison of Endoscopic Retrograde Cholangiopancreatography in Pediatric Patients: A 10-year Experience. *J Pediatr Gastroenterol Nutr*. 2020;70(5):568-73.
 27. Fishman DS, Barth B, Man-Wai Tsai C, et al. A prospective multicenter analysis from the Pediatric ERCP Database Initiative: predictors of choledocholithiasis at ERCP in pediatric patients. *Gastrointest Endosc*. 2021;94(2):311-7.e1.
 28. Liu QY, Ruan W, Fishman DS, et al. Predictors of Prolonged Fluoroscopy Exposure in Pediatric Endoscopic Retrograde Cholangiopancreatography: Results From the Large Pediatric Endoscopic Retrograde Cholangiopancreatography Database Initiative Multicenter Cohort. *J Pediatr Gastroenterol Nutr*. 2022;74(3):408-12.
 29. Dillman JR, Patel RM, Lin TK, Towbin AJ, Trout AT. Diagnostic performance of magnetic resonance cholangiopancreatography (MRCP) versus endoscopic retrograde cholangiopancreatography (ERCP) in the pediatric population: a clinical effectiveness study. *Abdom Radiol (NY)*. 2019;44(7):2377-83.
 30. Troendle DM, Fishman DS, Barth BA, et al. Therapeutic Endoscopic Retrograde Cholangiopancreatography in Pediatric Patients With Acute Recurrent and Chronic Pancreatitis: Data From the INSPPIRE (INternational Study group of Pediatric Pancreatitis: In search for a cuRE) Study. *Pancreas*. 2017;46(6):764-9.
 31. Tagawa M, Morita A, Imagawa K, Mizokami Y. Endoscopic retrograde cholangiopancreatography and endoscopic ultrasound in children. *Dig Endosc*. 2021;33(7):1045-58.
 32. Asenov Y, Akin M, Cantez S, Gün Soysal F, Tekant Y. Endoscopic retrograde cholangiopancreatography in children: Retrospective series with a long-term follow-up and literature review. *Turk J Gastroenterol*. 2019;30(2):192-7.
 33. Rosen JD, Lane RS, Martinez JM, et al. Success and safety of endoscopic retrograde cholangiopancreatography in children. *J Pediatr Surg*. 2017;52(7):1148-51.
 34. Zeng JQ, Deng ZH, Yang KH, et al. Endoscopic retrograde cholangiopancreatography in children with symptomatic pancreaticobiliary maljunction: A retrospective multicenter study. *World J Gastroenterol*. 2019;25(40):6107-15.
 35. Vitale DS, Lin TK. Trends in Pediatric Endoscopic Retrograde Cholangiopancreatography and Interventional Endoscopy. *J Pediatr*. 2021;232:10-2.

36. Makita S, Amano H, Kawashima H, et al. Utility of endoscopic retrograde cholangiopancreatography in management of pediatric pancreaticobiliary disease. *BMC Pediatr.* 2022;22(1):134.
37. Hassan AM, Lin TK, Smith MT, Hornung L, Abu-El-Haija M, Nathan JD, Vitale DS. Risk Factors for Post-ERCP Pancreatitis in Pediatric and Young Adult Patients. *J Pediatr Gastroenterol Nutr.* 2023;76(6):807-12.
38. Gupta R, Khalaf RT, Morrison J, et al. Impact of Trainee Involvement on Pediatric ERCP Procedures: Results From the Pediatric ERCP Initiative. *J Pediatr Gastroenterol Nutr.* 2023;77(1):126-30.
39. Garg L, Vaidya A, Kale A, Gopan A, Ansari A, Patra BR, Shukla A. Safety and efficacy of endoscopic retrograde cholangiopancreatography in pediatric pancreatic and biliary disorders. *Indian J Gastroenterol.* 2024;43(5):1037-44.
40. Eldredge RS, Notrica DM, Patel M, Garvey E, Ochoa B, McOmber M. Use of Endoscopic Retrograde Cholangiopancreatography in the Diagnosis and Management of Pediatric Pancreatic Injury. *J Pediatr Surg.* 2025;60(1):161962.
41. Overman RE, Hsieh LB, Thomas TT, Gadepalli SK, Geiger J. Pediatric Laparoscopic Common Bile Duct Exploration: An Opportunity to Decrease ERCP Complications. *J Surg Res.* 2019;242:318-22.
42. Hatlani MA, Kortan P, May G, Ling SC, Walters T, Avitzur Y. Wire-guided cannulation versus contrast-guided cannulation in pediatric endoscopic retrograde cholangiopancreatography. *Saudi J Gastroenterol.* 2015;21(1):25-9.
43. Joseph M, Schiff R, Mark J, Kramer R. Influence of pediatric ERCP positioning on procedural outcomes: A single-center study. *J Pediatr Gastroenterol Nutr.* 2025;80(2):345-52.
44. Wen J, Li T, Liu L, Bie LK, Gong B. Long-term outcomes of therapeutic ERCP in pediatric patients with pancreas divisum presenting with acute recurrent or chronic pancreatitis. *Pancreatology.* 2019;19(6):834-41.
45. Lin TK, Pathak SJ, Hornung LN, Vitale DS, Nathan JD, Abu-El-Haija M. Clinical Outcomes Following Therapeutic Endoscopic Retrograde Cholangiopancreatography in Children With Pancreas Divisum. *J Pediatr Gastroenterol Nutr.* 2021;72(2):300-5.
46. Zhang J, Gao LC, Guo S, et al. Endoscopic retrograde cholangiopancreatography in the treatment of pancreaticopleural fistula in children. *World J Gastroenterol.* 2020;26(37):5718-30.
47. Pathak SJ, Attard T, Hall M, Arain M, Heyman MB, Perito ER. Availability and utilization of endoscopic retrograde cholangiopancreatography at children's hospitals. *J Pediatr Gastroenterol Nutr.* 2024;78(5):1180-9.
48. Hiramatsu T, Itoh A, Kawashima H, et al. Usefulness and safety of endoscopic retrograde cholangiopancreatography in children with pancreaticobiliary maljunction. *J Pediatr Surg.* 2015;50(3):377-81.
49. Qian M, Wang J, Sun S, et al. Efficacy and Safety of Endoscopic Retrograde Cholangiopancreatography in Children of Pancreaticobiliary Maljunction Without Obvious Biliary Dilatation. *J Pediatr Surg.* 2024;59(4):653-9.
50. Kieling CO, Hallal C, Spessato CO, Ribeiro LM, Breyer H, Goldani HA, Maguilnik I. Changing pattern of indications of endoscopic retrograde cholangiopancreatography in children and adolescents: a twelve-year experience. *World J Pediatr.* 2015;11(2):154-9.
51. Barakat MT, Gugig R, Imperial J, Berquist WE. Fluoroscopy Time During Endoscopic Retrograde Cholangiopancreatography Performed for Children and Adolescents is Significantly Higher With Low-volume Endoscopists. *J Pediatr Gastroenterol Nutr.* 2021;72(2):244-9.
52. Oracz G, Pertkiewicz J, Dadalski M, Lipinski P, Wojno W, Tesseire M, Woynarowski M. Complications of endoscopic retrograde cholangiopancreatography in children with chronic pancreatitis depends on center experience. *J Pediatr.* 2017;186:215-6.
53. Wang Q, Moon SB, Zang J, Liu J, Weng H, Wang X, Wang J, Chen J. Usefulness of pre-operative endoscopic retrograde cholangiopancreatography in diagnosis and management of forme fruste choledochal cyst in children. *ANZ J Surg.* 2020;90(6):1041-5.
54. Mark JA, Kramer RE. Ketorolac Is Safe and Associated With Lower Rate of Post-Endoscopic Retrograde Cholangiopancreatography Pancreatitis in Children With Pancreatic Duct Manipulation. *J Pediatr Gastroenterol Nutr.* 2021;73(4):542-7.
55. Zhang W, Hou SL, Hou YK, Tian J, Zhang LC. Transduodenal accessory papillary rendezvous technique assisted ERCP for the treatment of severe pancreatic duct stenosis caused by chronic pancreatitis in children. *Rev Esp Enferm Dig.* 2024;116(12):718-9.
56. Dahale AS, Puri AS, Sachdeva S, Srivastava S, Kumar A. Endoscopic Retrograde Cholangiopancreatography in Children: A Single-center Experience From Northern India. *Indian Pediatr.* 2019;56(3):196-8.
57. Li Q, Li S, Hou S, et al. ERCP-Related adverse events in pediatric patients: a 10-years single-site review. *Pediatr Surg Int.* 2024;40(1):199.
58. Cho JM, Jeong IS, Kim HJ, Oh SH, Kim KM. Early adverse events and long-term outcomes of endoscopic sphincterotomy in a pediatric population: a single-center experience. *Endoscopy.* 2017;49(5):438-46.
59. Elmunzer BJ, Foster LD, Serrano J, et al. Indomethacin with or without prophylactic pancreatic stent placement to prevent pancreatitis after ERCP: a randomised non-inferiority trial. *Lancet.* 2024;403(10425):450-8.
60. Perera K, Nawarathne N, Samarawickrama VT, Deraniyagala MP, Luxman W, Fernandopulle A. Endoscopic Retrograde Cholangiopancreatography in Children: Feasibility, Success, and Safety with Standard Adult Endoscopes and Accessories. *Pediatr Gastroenterol Hepatol Nutr.* 2022;25(5):406-12.
61. Troendle D, Abraham O, Huang R, Barth Ba. 440 Post-ERCP Pancreatitis in Children and Adolescents. *GASTROINTESTINAL ENDOSCOPY.* 2014;79(5):AB140-140AB141.
62. Abu-El-Haija M, Kumar S, Szabo F, Werlin S, Conwell D, Banks P, Morinville VD, NASPGHAN Pancreas Committee.

- Classification of Acute Pancreatitis in the Pediatric Population: Clinical Report From the NASPGHAN Pancreas Committee. *J Pediatr Gastroenterol Nutr.* 2017;64(6):984-90.
63. Batıbay E, Yüsekayla O, Polat M, et al. Endoscopic Retrograde Cholangiopancreatography in Pediatric Population: A Decade-Long Experience from 2 Tertiary Centers. *Turk J Gastroenterol.* 2025;36(5):321-7.
 64. Coté CJ, Wilson S, AMERICAN ACADEMY OF PEDIATRICS, AMERICAN ACADEMY OF PEDIATRIC DENTISTRY. Guidelines for Monitoring and Management of Pediatric Patients Before, During, and After Sedation for Diagnostic and Therapeutic Procedures: Update 2016. *Pediatrics.* 2016;138(1).
 65. Troendle DM, Abraham O, Huang R, Barth BA. Factors associated with post-ERCP pancreatitis and the effect of pancreatic duct stenting in a pediatric population. *GASTROINTESTINAL ENDOSCOPY.* 2015;81(6):1408-16.
 66. Cheng CL, Fogel EL, Sherman S, et al. Diagnostic and therapeutic endoscopic retrograde cholangiopancreatography in children: a large series report. *J Pediatr Gastroenterol Nutr.* 2005;41(4):445-53.
 67. Suzuki M, Minowa K, Nakano S, Isayama H, Shimizu T. Genetic Abnormalities in Pancreatitis: An Update on Diagnosis, Clinical Features, and Treatment. *Diagnosics (Basel).* 2020;11(1).
 68. Cotton PB, Garrow DA, Gallagher J, Romagnuolo J. Risk factors for complications after ERCP: a multivariate analysis of 11,497 procedures over 12 years. *Gastrointest Endosc.* 2009;70(1):80-8.
 69. Testoni PA, Mariani A, Giussani A, et al. Risk factors for post-ERCP pancreatitis in high- and low-volume centers and among expert and non-expert operators: a prospective multicenter study. *Am J Gastroenterol.* 2010;105(8):1753-61.
 70. Kohoutova D, Tringali A, Papparella G, Perri V, Boškoski I, Hamanaka J, Costamagna G. Endoscopic treatment of chronic pancreatitis in pediatric population: Long-term efficacy and safety. *United European Gastroenterol J.* 2019;7(2):270-7.
 71. Troendle DM, Gurrin B, Huang R, Barth BA. IV Ibuprofen for Prevention of Post-ERCP Pancreatitis in Children: A Randomized Placebo-controlled Feasibility Study. *J Pediatr Gastroenterol Nutr.* 2020;70(1):121-6.
 72. Goetz M, Andersen P, Bergman J, Frei N, Schmidt A, Kähler G, Martus P, Dechêne A. ERCP in babies: Low risk of post-ERCP pancreatitis - results from a multicentre survey. *United European Gastroenterol J.* 2020;8(1):77-80.
 73. Akshintala VS, Kanthasamy K, Bhullar FA, et al. Incidence, severity, and mortality of post-ERCP pancreatitis: an updated systematic review and meta-analysis of 145 randomized controlled trials. *Gastrointest Endosc.* 2023;98(1):1-6.e12.
 74. Freeman ML, DiSario JA, Nelson DB, et al. Risk factors for post-ERCP pancreatitis: a prospective, multicenter study. *Gastrointest Endosc.* 2001;54(4):425-34.
 75. Cheng CL, Sherman S, Watkins JL, et al. Risk factors for post-ERCP pancreatitis: a prospective multicenter study. *Am J Gastroenterol.* 2006;101(1):139-47.
 76. Testoni PA, Mariani A, Giussani A, et al. Risk factors for post-ERCP pancreatitis in high- and low-volume centers and among expert and non-expert operators: a prospective multicenter study. *Am J Gastroenterol.* 2010;105(8):1753-61.
 77. Buxbaum JL, Freeman M, Amateau SK, et al. American Society for Gastrointestinal Endoscopy guideline on post-ERCP pancreatitis prevention strategies: summary and recommendations. *Gastrointest Endosc.* 2023;97(2):153-62.
 78. Luo H, Zhao L, Leung J, et al. Routine pre-procedural rectal indometacin versus selective post-procedural rectal indometacin to prevent pancreatitis in patients undergoing endoscopic retrograde cholangiopancreatography: a multicentre, single-blinded, randomised controlled trial. *Lancet.* 2016;387(10035):2293-301.
 79. Zuo J, Li H, Zhang S, Li P. Nonsteroidal Anti-inflammatory Drugs for the Prevention of Post-endoscopic Retrograde Cholangiopancreatography Pancreatitis. *Dig Dis Sci.* 2024;69(9):3134-46.
 80. Patel R, Bertran-Rodriguez C, Kumar A, Brady P, Gomez-Esquivel R, Changela K, Niknam N, Taunk P. Efficacy of aggressive hydration with normal saline versus lactated Ringer's solution for the prevention of post-ERCP pancreatitis in high-risk patients: a randomized controlled trial. *Endosc Int Open.* 2022;10(7):E933-933E939.
 81. Severino A, Varca S, Airola C, Mezza T, Gasbarrini A, Franceschi F, Candelli M, Nista EC. Antibiotic Utilization in Acute Pancreatitis: A Narrative Review. *Antibiotics (Basel).* 2023;12(7).
 82. Ferreira LE, Baron TH. Post-sphincterotomy bleeding: who, what, when, and how. *Am J Gastroenterol.* 2007;102(12):2850-8.
 83. Shin JH. Refractory gastrointestinal bleeding: role of angiographic intervention. *Clin Endosc.* 2013;46(5):486-91.
 84. Theopistos V, Theocharis G, Konstantakis C, Kitrou P, Kehagias I, Triantos C, Thomopoulos K. Non-Operative Management of Type 2 ERCP-Related Retroperitoneal Duodenal Perforations: A 9-Year Experience From a Single Center. *Gastroenterology Res.* 2018;11(3):207-12.
 85. Paspatis GA, Arvanitakis M, Dumonceau JM, et al. Diagnosis and management of iatrogenic endoscopic perforations: European Society of Gastrointestinal Endoscopy (ESGE) Position Statement - Update 2020. *Endoscopy.* 2020;52(9):792-810.
 86. Pal P, Ramchandani M. Management of ERCP complications. *Best Pract Res Clin Gastroenterol.* 2024;69:101897.
 87. Gomi H, Solomkin JS, Schlossberg D, et al. Tokyo Guidelines 2018: antimicrobial therapy for acute cholangitis and cholecystitis. *J Hepatobiliary Pancreat Sci.* 2018;25(1):3-16.
 88. Cotton PB, Lehman G, Vennes J, Geenen JE, Russell RC, Meyers WC, Liguory C, Nickl N. Endoscopic sphincterotomy complications and their management: an attempt at consensus. *Gastrointest Endosc.* 1991;37(3):383-93.
 89. Smeets X, Bouhouch N, Buxbaum J, et al. The revised Atlanta criteria more accurately reflect severity of post-ERCP pancreatitis compared to the consensus criteria. *United European Gastroenterol J.* 2019;7(4):557-64.

Additional files

Additional file 1. PRISMA 2020 checklist. DOCX. Completed PRISMA 2020 checklist for the systematic review and meta-analysis.

Additional file 2. PRISMA flowchart. DOCX. Document version of the study identification, screening, eligibility,

and inclusion flowchart.

Additional file 3. Reconstructed study-level extraction sheet. CSV. Study-level extraction sheet reconstructed from the manuscript, source tables, and final verification files.

Additional file 4. Verified PEP source dataset. CSV. Verified event counts, denominators, and source notes for the PEP meta-analysis.

Additional file 5. PEP calculated values. CSV. Study-level proportions, confidence intervals, transformed values, standard errors, and random-effects weights.

Additional file 6. PEP meta-analysis summary. CSV. Pooled PEP estimate, confidence interval, heterogeneity statistics, and model-scale values.

Additional file 7. Exploratory Egger regression output. CSV. Approximate Egger regression output for the exploratory PEP funnel-plot assessment.

Additional file 8. PEP source-data verification report. DOCX. Audit trail for the corrected Goetz 2020 and Mercier 2021 rows and verified Perera 2022 and Batibay 2025 rows.

Additional file 9. PEP analysis and figure-generation script. TXT. Plain-text copy of the PowerShell script used to regenerate the PEP forest plot, funnel plot, pooled estimate, and supporting CSV outputs.

Figures and tables

Figure 1. PRISMA 2020 flow diagram showing the selection process of eligible studies. A total of 1093 records were identified from four databases (PubMed, Embase, Web of Science, and Cochrane Library), and 65 sources were included in the qualitative synthesis.

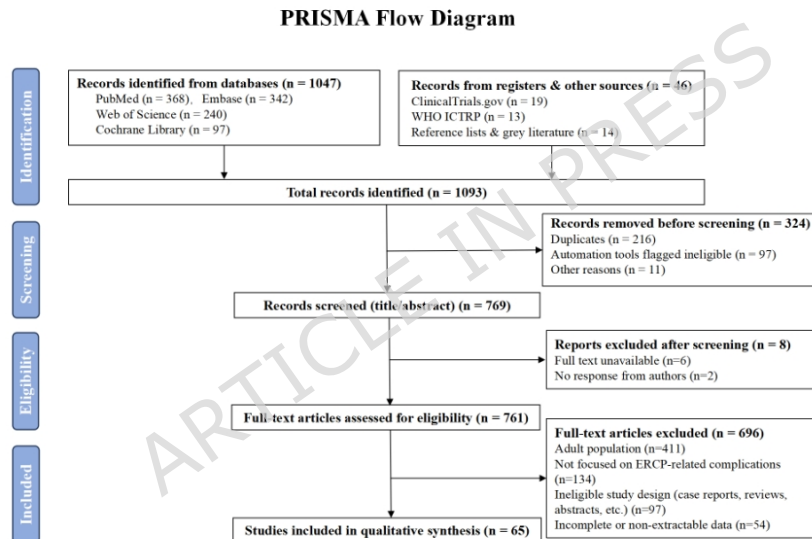
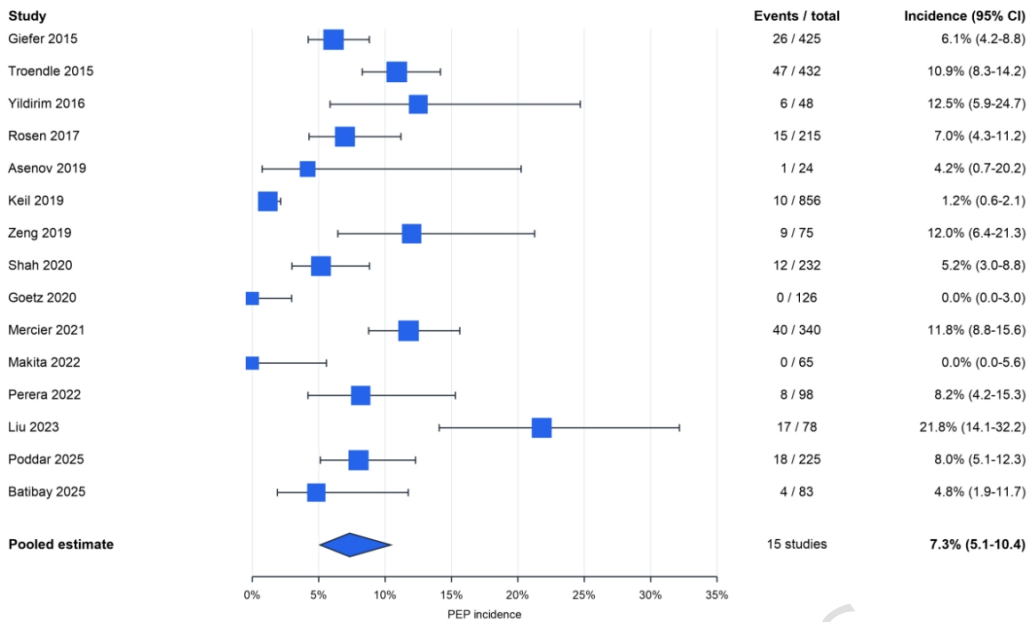


Figure 2. Forest plot of PEP incidence after pediatric ERCP. Random-effects meta-analysis of logit-transformed proportions across 15 non-overlapping primary studies with extractable PEP numerators and denominators. The pooled PEP incidence was 7.3% (95% CI, 5.1% to 10.4%; I-squared = 83.7%). Denominators follow the reporting unit used in each source, and source-level verification notes are provided in the extraction dataset.

Figure 2. Forest plot of PEP incidence after pediatric ERCP

Random-effects meta-analysis of logit-transformed proportions



Heterogeneity: $Q = 85.85$, $I^2 = 83.7\%$, $\tau^2 = 0.412$. Denominators follow each source report.
 PEP, post-ERCP pancreatitis; ERCP, endoscopic retrograde cholangiopancreatography.

Figure 3. Evidence map of risk domains for post-ERCP complications in children. This figure compares patient-related and procedure-related risk domains reported across four major post-ERCP complication categories. Procedure-related factors, particularly pancreatic duct manipulation and sphincterotomy, accounted for most recurring risk signals, whereas patient-level factors, including younger age and prior pancreatitis, also contributed to risk.

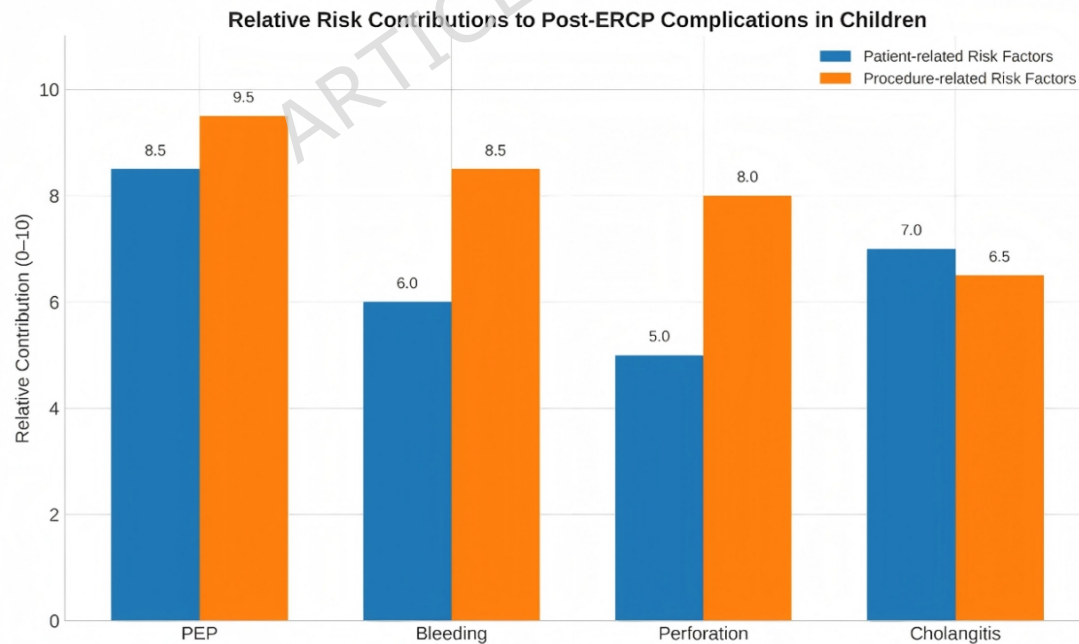


Figure 4. Age-stratified incidence of PEP in pediatric patients. This bar graph illustrates the estimated incidence of PEP across pediatric age groups and should be interpreted as hypothesis-generating rather than definitive.

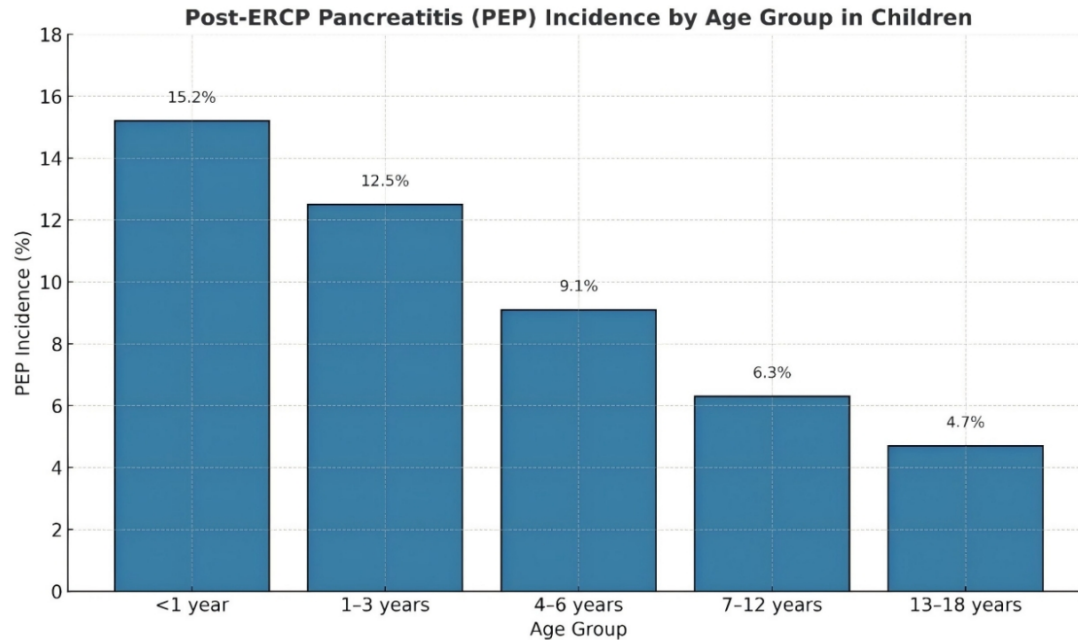
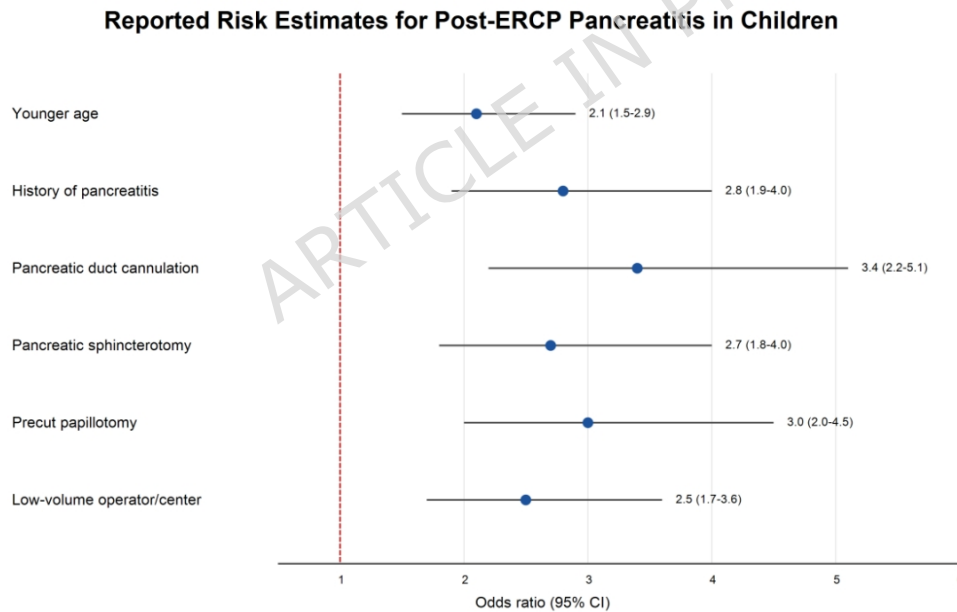


Figure 5. Reported effect estimates for selected PEP risk factors in children. Reported odds ratios with 95% confidence intervals are shown for key predictors where extractable estimates were available. Because covariate adjustment varied across source studies, this visual summary is descriptive rather than a pooled predictor meta-analysis.



Descriptive evidence map based on extractable reported estimates; not a de novo pooled meta-analysis.

Figure 6. Summary of prevention strategies for PEP in children. This horizontal bar chart summarizes prophylactic approaches according to clinical utility and maturity of supporting evidence. Experienced operator involvement and minimized pancreatic manipulation received the highest recommendation levels, whereas rectal NSAIDs and intravenous hydration still require stronger pediatric validation.

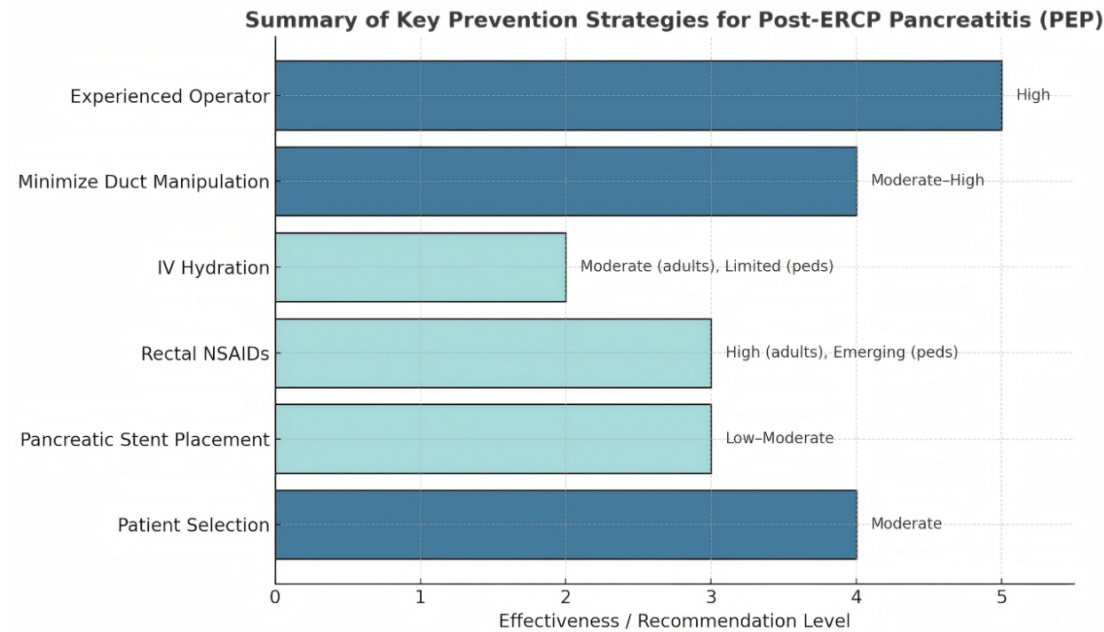
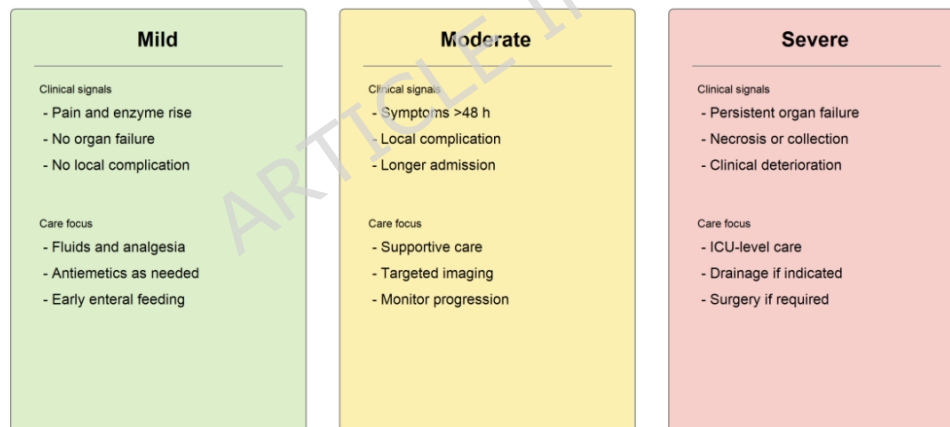


Figure 7. Pragmatic severity and management framework for PEP in children. A simplified framework for PEP severity and stepwise management in children, adapted from revised Atlanta concepts and pediatric practice considerations. It should be used as a practical clinical framework rather than a validated pediatric severity score.

PEP Severity and Escalation Framework in Children

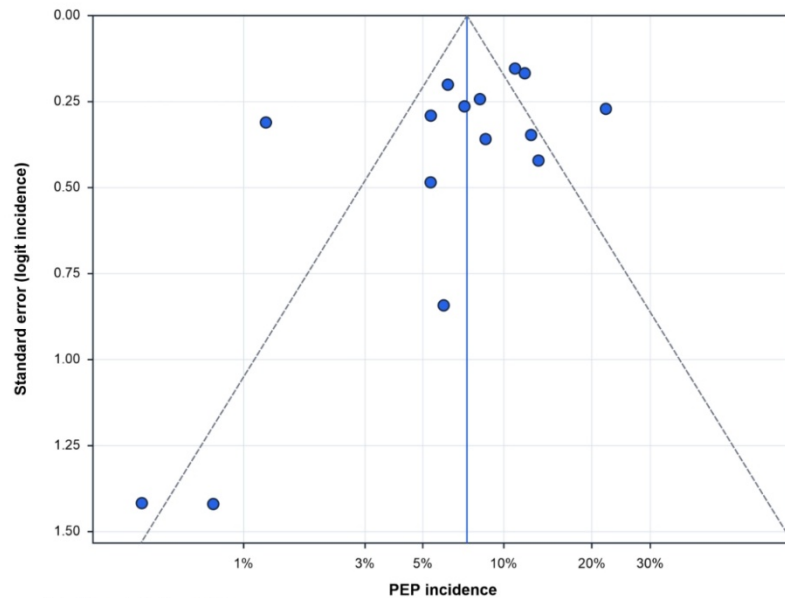


Adapted from adult Atlanta concepts; pediatric-specific validation remains limited.
Use as a pragmatic escalation framework, not as a validated pediatric severity score.

Figure 8. Funnel plot for PEP incidence after pediatric ERCP. Exploratory funnel plot based on logit-transformed PEP proportions from studies included in Figure 2. The plot screens for small-study effects and should be interpreted cautiously because of substantial heterogeneity, rare-event behavior, and mixed patient/procedure denominators.

Figure 8. Funnel plot for PEP incidence

Exploratory small-study-effect assessment on the logit scale



Vertical line: pooled incidence 7.3%.

Dashed lines show approximate 95% pseudo-confidence limits; interpretation is limited by heterogeneity.

Table 1. Primary PEP studies with percentages, denominator units, and definitions.

Source	Source type	PEP n/N (%)	Denominator unit	Definition or reporting note	Role
Giefer 2015	Primary pediatric ERCP study	26/425 (6.1%)	ERCPs	PEP reported after ERCP; event count from PubMed abstract.	Included in PEP meta-analysis
Troendle 2015	Primary pediatric ERCP study	47/432 (10.9%)	ERCPs	PEP reported after pediatric ERCP; pancreatic duct stenting evaluated.	Included in PEP meta-analysis
Yildirim 2016	Primary pediatric ERCP study	6/48 (12.5%)	Patients	PEP count and denominator extracted from the source record used in the prior review table; reported at patient level.	Included in PEP meta-analysis
Rosen 2017	Primary pediatric ERCP study	15/215 (7.0%)	ERCPs or patients reported	PEP count as reported in the study record; denominator unit could not be further harmonized.	Included in PEP meta-analysis
Asenov 2019	Primary pediatric ERCP study	1/24 (4.2%)	Patients	Mild post-procedure pancreatitis reported among 24 patients.	Included in PEP meta-analysis
Keil 2019	Primary pediatric ERCP study	10/856 (1.2%)	ERCPs	Pancreatitis reported after ERCP interventions in the full text.	Included in PEP meta-analysis
Zeng 2019	Primary pediatric ERCP study	9/75 (12.0%)	Patients	PEP count and denominator extracted from the source record used in the prior review table.	Included in PEP meta-analysis
Shah 2020	Primary pediatric ERCP study	12/232 (5.2%)	Mixed unclear or	PEP count from source record; denominator reported as pediatric ERCPs/patients and retained as the source unit.	Included in PEP meta-analysis

Goetz 2020	Primary pediatric ERCP study	0/126 (0.0%)	First ERCP per child	Full-text source reported no PEP among 126 infants undergoing first ERCP.	Included in PEP meta-analysis
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Table 2. Risk factors for major post-ERCP complications in children.

Complication	Patient-related factors	Procedure-related factors	Interpretation
PEP	Younger age, prior acute recurrent or chronic pancreatitis, suspected sphincter of Oddi dysfunction, hereditary pancreatitis variants	Repeated or deep pancreatic duct cannulation, pancreatic sphincterotomy, precut papillotomy, high-volume pancreatic contrast injection, prolonged procedure, lower operator or center volume	Most consistently reported complication; definitions and adjustment strategies vary across studies
Bleeding	Coagulopathy, antiplatelet or anticoagulant exposure, portal hypertension, active inflammation	Endoscopic sphincterotomy, precut papillotomy, large stone extraction, repeat intervention	Usually uncommon and endoscopically manageable, but delayed bleeding should be monitored
Perforation	Altered anatomy, strictures, severe inflammation or fibrosis, small pediatric anatomy	Difficult cannulation, excessive guidewire or endoscope force, balloon dilation, stent placement, precut papillotomy	Rare but clinically severe; early recognition determines management
Cholangitis	Biliary obstruction, prior biliary surgery	Incomplete drainage, incomplete stone clearance	Prevention depends on adequate drainage and

	immunocompromised status, biliary stents	contaminated instrument, prolonged procedure	selective antibiotics
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Table 3. Prevention and management strategies for post-ERCP complications in children.

Complication	Prevention	Management
PEP	Risk stratification by age, indication, prior pancreatitis, and anticipated pancreatic manipulation; minimize pancreatic duct cannulation and contrast injection; selective prophylactic pancreatic stenting after high-risk pancreatic instrumentation with stent caliber and length matched to pancreatic duct size and child size; NSAID prophylaxis only after checking age, weight, renal function, bleeding risk, and local pediatric protocols; monitored hydration; experienced pediatric ERCP operator.	Conservative care with intravenous fluids, analgesia, antiemetics, and early enteral nutrition when feasible; intensive monitoring for systemic inflammatory response; drainage or debridement only for selected infected necrosis or persistent collections.
Bleeding	Correct coagulopathy or thrombocytopenia before ERCP when possible; review antiplatelet or anticoagulant exposure; use careful sphincterotomy or papillotomy technique; consider selective prophylactic hemostasis in high-risk lesions.	Endoscopic injection, clipping, or thermal coagulation; transfusion when unstable; angiographic embolization or surgery for refractory cases.
Perforation	Meticulous cannulation, avoidance of excessive guidewire or endoscope force, recognition of altered anatomy, and early senior-operator takeover during difficult access.	Conservative care for selected small retroperitoneal perforations; endoscopic closure in selected cases; surgical repair for large defects, free perforation, or peritonitis.
Cholangitis	Confirm adequate biliary drainage; use selective antibiotics when drainage is incomplete, active cholangitis is present, the patient is immunocompromised, or biliary stents remain in place; maintain sterile technique.	Broad-spectrum antibiotics and restoration of biliary drainage by ERCP, percutaneous intervention, or surgery when needed.

Table 4. Comparison of the present PEP synthesis with prior pediatric ERCP meta-analyses.

Study or synthesis	Scope	Reported estimate	Key methodological difference from the present review	Interpretation
Usatin 2016 [13]	Earlier pediatric ERCP literature	PEP about 3% (95% CI, 2% to 5%)	Older practice era and broad inclusion; patient/procedure units were variably reported.	Lower estimate likely reflects earlier cohorts, mixed indications, and different reporting rules.
Sun 2022 [5]	Therapeutic pediatric ERCP	Overall post-ERCP adverse events about 8% (95% CI, 6% to 10%); PEP was not always separately extractable.	Therapeutic-only focus and pooled adverse events rather than a PEP-specific synthesis for every source.	Useful safety benchmark, but not directly comparable with a PEP-only meta-analysis.
Hosseini 2023 [4]	Pediatric ERCP indications, success, and adverse events	PEP about 4% (95% CI, 3% to 6%)	Larger historical evidence base with broader inclusion and possible differences in cohort-overlap handling.	Lower estimate than the present review may reflect broader historical sampling and denominator heterogeneity.
Present review	Contemporary	Random-effects PEP	Restricted to non-overlapping	Higher estimate probably

	pediatric ERCP studies from Jan 2015 to Jun 2025	incidence 7.3% (95% CI, 5.1% to 10.4%; I-squared = 83.7%).	primary studies with extractable PEP counts; denominator unit retained and displayed.	reflects contemporary high-risk therapeutic practice, pancreatic interventions, explicit PEP extraction, and definition/reporting heterogeneity.
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PRISMA Flow Diagram

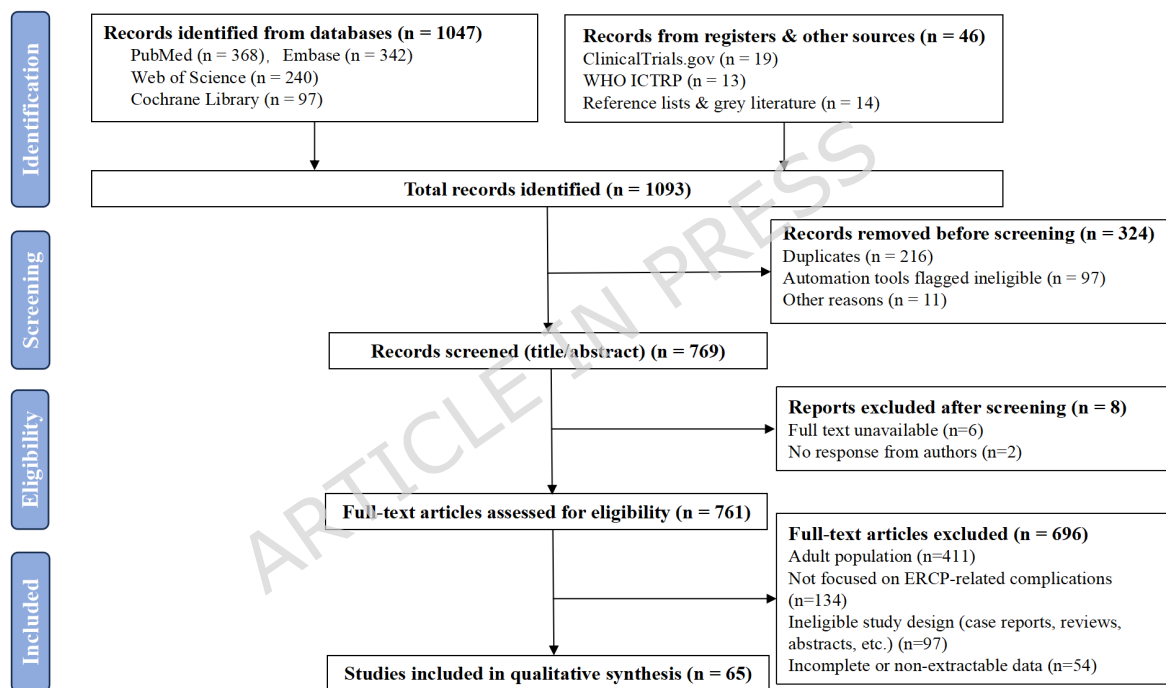
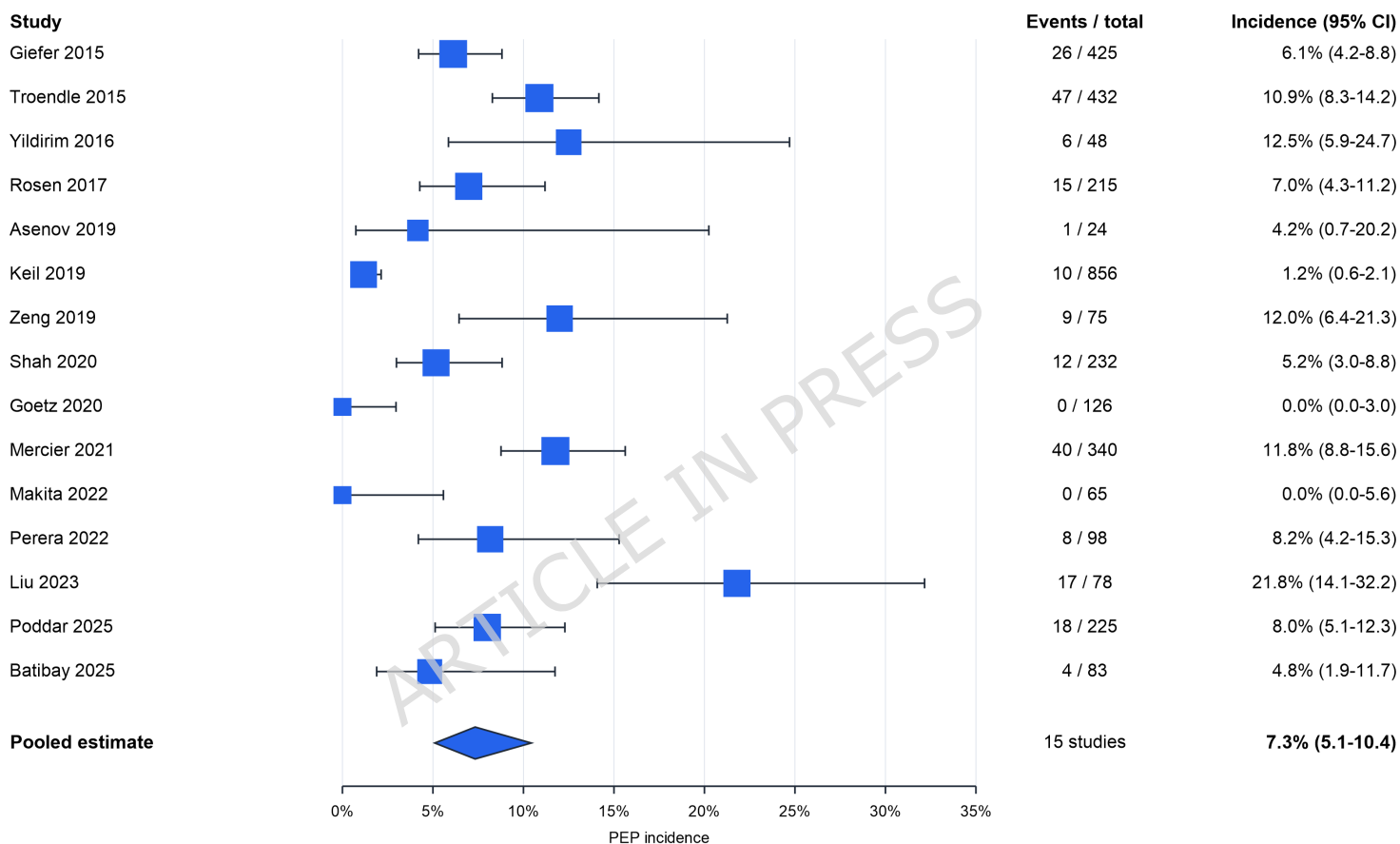
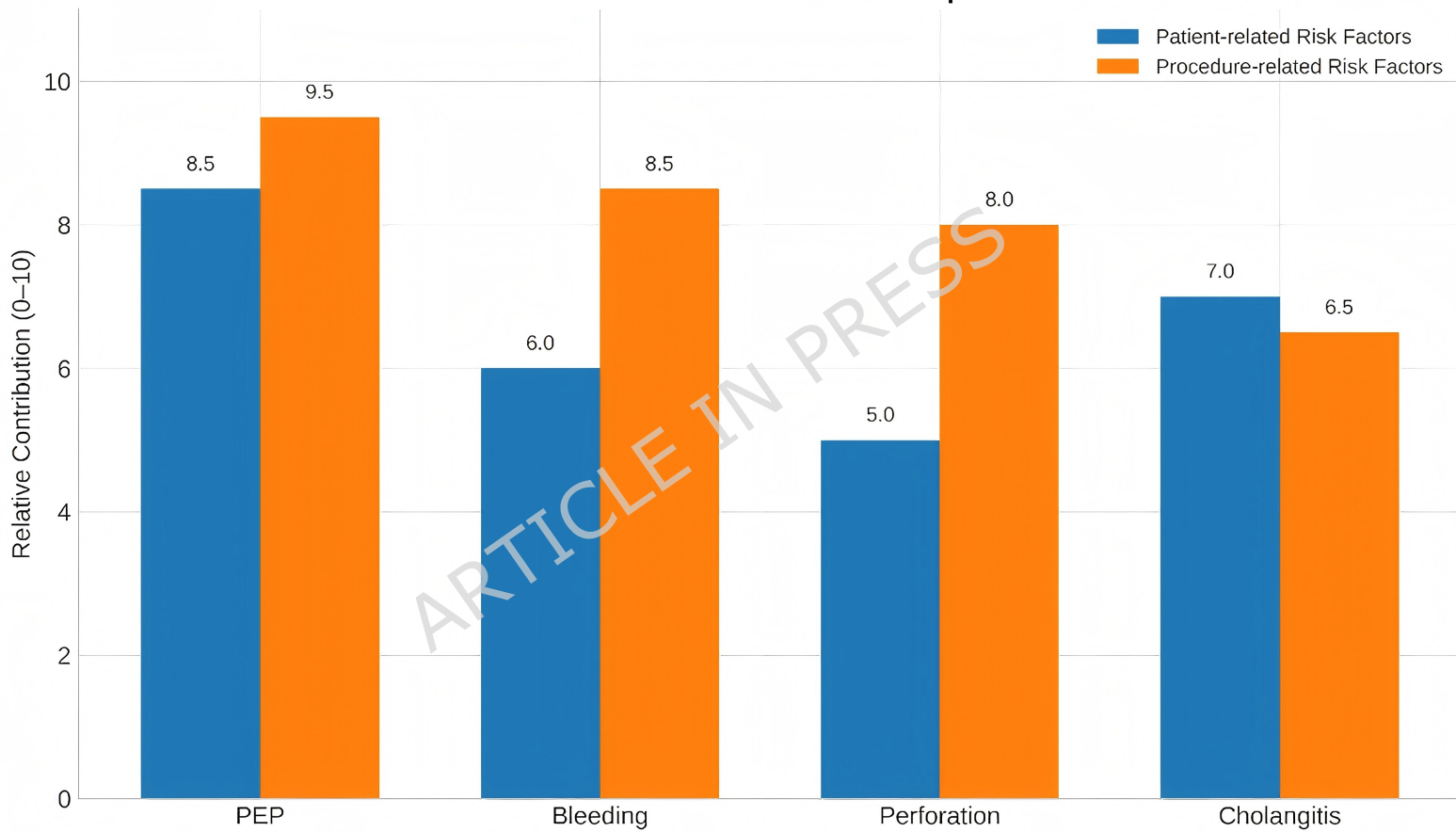


Figure 2. Forest plot of PEP incidence after pediatric ERCP

Random-effects meta-analysis of logit-transformed proportions

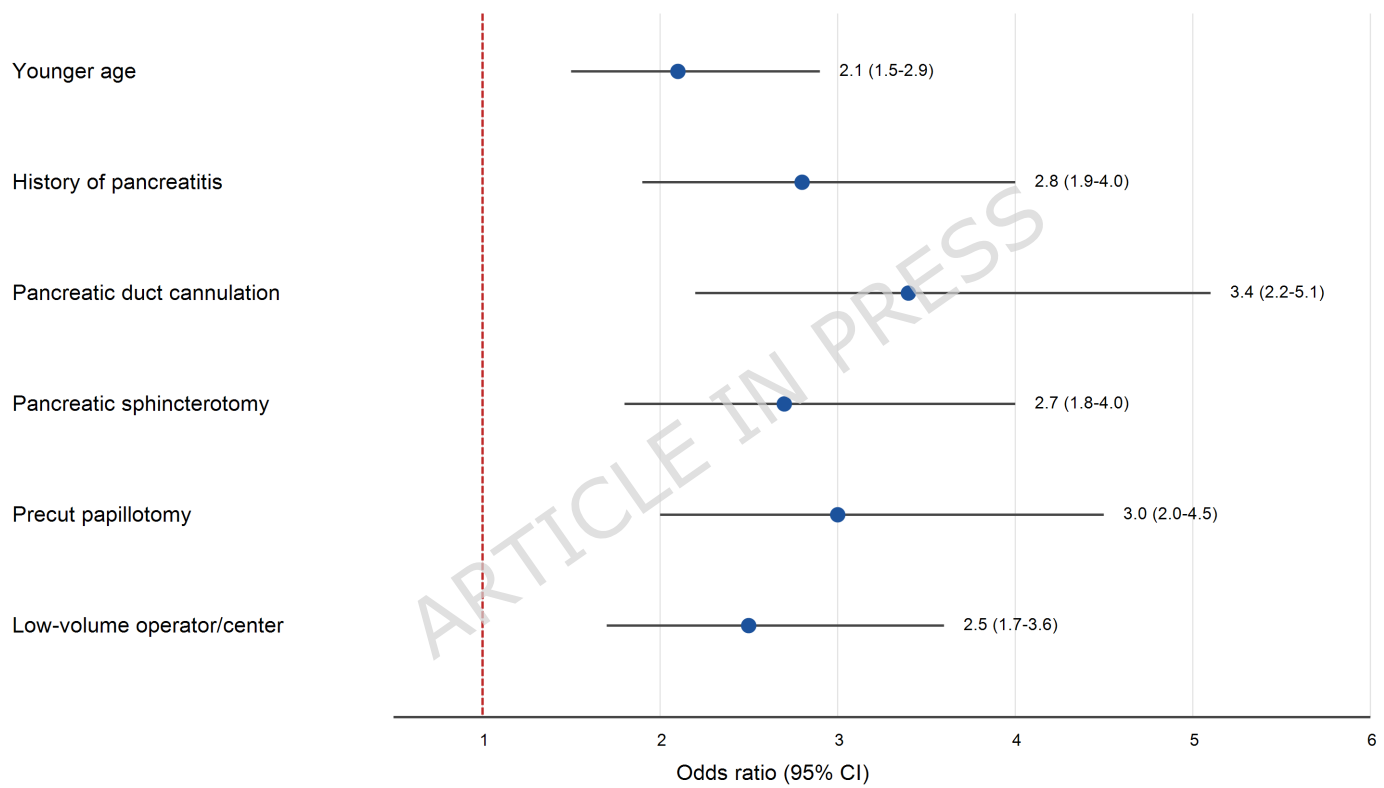
Heterogeneity: $Q = 85.85$, $I^2 = 83.7\%$, $\tau^2 = 0.412$. Denominators follow each source report.

PEP, post-ERCP pancreatitis; ERCP, endoscopic retrograde cholangiopancreatography.

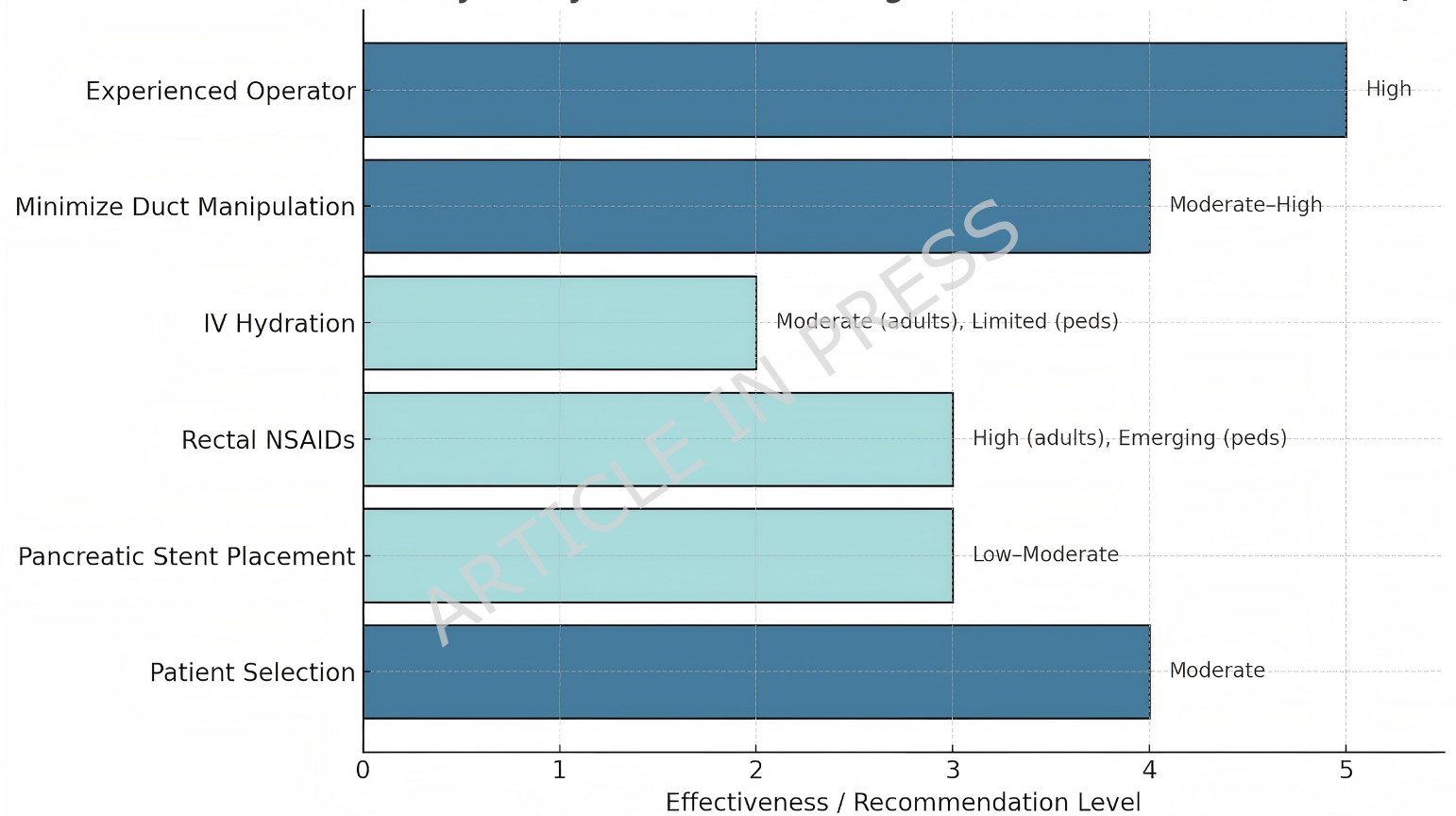
Relative Risk Contributions to Post-ERCP Complications in Children

Post-ERCP Pancreatitis (PEP) Incidence by Age Group in Children

Reported Risk Estimates for Post-ERCP Pancreatitis in Children



Descriptive evidence map based on extractable reported estimates; not a de novo pooled meta-analysis.

Summary of Key Prevention Strategies for Post-ERCP Pancreatitis (PEP)

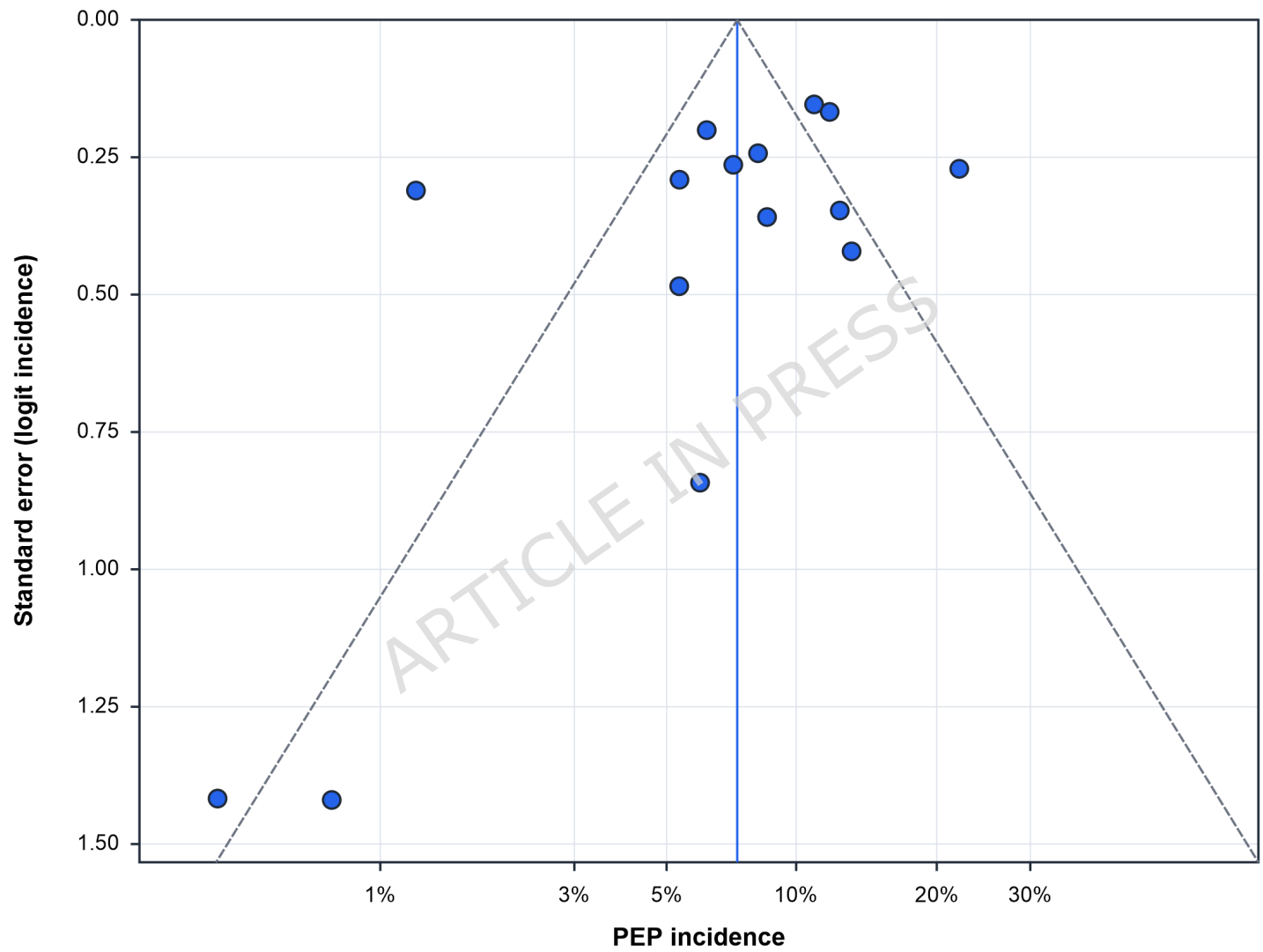
PEP Severity and Escalation Framework in Children

Mild	Moderate	Severe
Clinical signals - Pain and enzyme rise - No organ failure - No local complication Care focus - Fluids and analgesia - Antiemetics as needed - Early enteral feeding	Clinical signals - Symptoms >48 h - Local complication - Longer admission Care focus - Supportive care - Targeted imaging - Monitor progression	Clinical signals - Persistent organ failure - Necrosis or collection - Clinical deterioration Care focus - ICU-level care - Drainage if indicated - Surgery if required

Adapted from adult Atlanta concepts; pediatric-specific validation remains limited.
Use as a pragmatic escalation framework, not as a validated pediatric severity score.

Figure 8. Funnel plot for PEP incidence

Exploratory small-study-effect assessment on the logit scale



Vertical line: pooled incidence 7.3%.

Dashed lines show approximate 95% pseudo-confidence limits; interpretation is limited by heterogeneity.

Table 1. Selected primary studies contributing to the PEP meta-analysis and contextual reviews

Source	Source type	PEP n/N (%)	Denominator unit	Definition or reporting note	Role
Giefer 2015	Primary pediatric ERCP study	26/425 (6.1%)	ERCPS	PEP reported after ERCP; event count from PubMed abstract.	Included in PEP meta-analysis
Troendle 2015	Primary pediatric ERCP study	47/432 (10.9%)	ERCPS	PEP reported after pediatric ERCP; pancreatic duct stenting evaluated.	Included in PEP meta-analysis
Yildirim 2016	Primary pediatric ERCP study	6/48 (12.5%)	Patients	PEP count and denominator extracted from the source record used in the prior review table; reported at patient level.	Included in PEP meta-analysis
Rosen 2017	Primary pediatric ERCP study	15/215 (7.0%)	ERCPS or patients as reported	PEP count as reported in the study record; denominator unit could not be further harmonized.	Included in PEP meta-analysis
Asenov 2019	Primary pediatric ERCP study	1/24 (4.2%)	Patients	Mild post-procedure pancreatitis reported among 24 patients.	Included in PEP meta-analysis
Keil 2019	Primary pediatric ERCP study	10/856 (1.2%)	ERCPS	Pancreatitis reported after ERCP interventions in the full text.	Included in PEP meta-analysis
Zeng 2019	Primary pediatric ERCP study	9/75 (12.0%)	Patients	PEP count and denominator extracted from the source record used in the prior review table.	Included in PEP meta-analysis
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